Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/02/2006
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Project Title:	Experimental and Numerical Investigations of Growth Morphologies of Peritectic Reactions		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
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Many engineering materials from steels to high temperature superconductors are produced from peritectic alloys, the properties and post-solidification processing of which are directly related their crystallization histories. The design of processes to control the microstructure and resultant properties of these materials requires knowledge of how the microstructure develops. Unfortunately, for peritectic alloys this development is not well understood. This study examines such microstructural growth arising from peritectic reactions, combining an experimental program using metal alloys and numerical predictions of microstructural morphology. This research outlines processing experiments and calculations which will include (a) a fundamental experimental study of the developing morphology of peritectic reactions and (b) the development of a complete model for the developing microstructure, using a cellular automata method and including nonequilibrium effects. The proposed study relies on a close coupling of these two thrusts to explain the morphological development of two-phase peritectic alloys.
The objectives for the research are to: (1) perform experiments with and without significant buoyancy driven convection

Task Description:	to discern the conditions under which morphological development of banded and coupled microstructures occurs in peritectic systems; (2) develop predictive models of such solidification behavior using cellular automaton methods to separate out physical mechanisms responsible for the morphologies; and (3) validate the numerical models by comparison with experiment. The overall impact of this work will be to obtain a greater understanding of the process-structure relationships for peritectic metal alloys. Not only can this work advance material science knowledge concerning these microstructures and their origins, it will be an opportunity to drive forward solidification modeling techniques. These models will be linked to macroscopic solidification codes, thereby increasing the span of length scales we can study simultaneously. The multiscale modeling developed in this project will be applicable to various industrial processes, including the solidification processing of grain-refined structural aluminum alloys and the refining of high temperature alloys for gas turbine engines used for power generation and aircraft propulsion. Strong, gravity-driven convective flows can disturb the distribution of solute in these processes, thereby having a significant impact on the development of the final microstructure. It is apparent that we must seek to limit the influence of gravity in order to develop the microstructural models and to understand the fundamental behavior of these alloys. It is also possible that the absence of buoyancy induced flows in microgravity could lead to easier processing of coupled microstructures in peritectic alloys.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A numerical model was developed to simulate solidification in a binary alloy using a modified cellular automaton. The model used here for the simulation of dendritic growth is a coupled cellular automata-finite volume procedure that can be divided into three parts. First, the cells to be examined are selected, then the growth or remelting of solid is predicted using the CA. Finally, the diffusion of solute is found using a finite volume method. Unlike previous CA models, the model developed at Purdue moves away from explicitly solving the interface mass balances to determine the velocity and position of the interface. In our approach, the velocity of the interface is controlled by the CA rules directly, responding to compositional changes away from equilibrium at the interface due to the diffusion of mass. The cells that may change include interface cells (containing both solid and liquid), liquid cells with at least one solid neighbor, and solid cells with at least one liquid neighbor. During the growth calculations, the cells with possible growth are polled in the random order and the phase changes that might occur are updated as soon as they are calculated. To find the fraction solid increase or decrease in a cell, it is necessary to calculate the equilibrium solute concentration (CL*) in the liquid at the solid-liquid interface, which is affected by both compositional and curvature undercooling. Both growth and remelting of the solid is simulated by the CA rules. As growth of the solid phase occurs, nonuniform compositional fields are formed in the liquid and solid due to the solute partitioning and rejection into the liquid. Subsequent solute diffusion in the liquid and the solid is predicted by the solution of the species conservation equation in two dimensions using an implicit finite volume method. As noted by several other researchers, there is an artificial, grid-oriented anisotropy in growth inherent to using a Cartesian grid because growth can only occur locally in the zero degree or 45 degree directions. Depending on the unmodified rule set used, we were able to simulate growth of dendrites oriented in one or the other of these two directions instead of in the specified growth direction. We have made successful modifications to the growth rates in the 0 or 45 degree directions to ameliorate this difficulty. We are now able to grow multiple dendrites with different orientations without the effect of this grid-induced anisotropy. Careful study of the behavior of the CA and its interaction with the diffusion model has led us to the prediction of ranges of grid spacing and time step that can be used in these simulations. These restrictions are inherent to the method, but little or no attention has been paid to them in previous studies. These ranges are functions of alloy properties and process parameters and are used to find numerical parameters that will give a stable and physically realistic morphology. We are now able to define process windows for grid spacing and time step for stable numerical solutions. The work is highlighted in a paper submitted to Modelling and Simulations in Materials Science and Engineering (July 2005) entitled " Modeling dendrites in Pb-Sn alloys using Cellular Automaton" by S. Raghavan, D. R. Johnson and M.J.M. Krane. The current work has been extended to model the solidification (grain structure and microsegregation) inside a preform during the manufacture of metal-matrix composites such as the application of abrasive tips for turbine blades.
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